WORKED SOLUTION

STEP II 2004, Question 10

Pure • Inclined plane

STEP II 2004, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q10

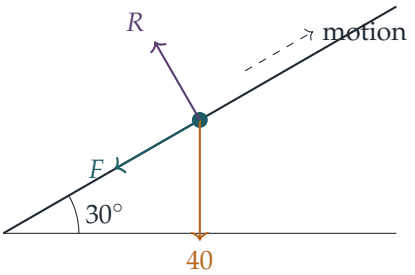
The key idea

Friction always opposes the motion, so a particle sliding up the plane has a different acceleration from one sliding down. Work out the two accelerations first: the numbers are chosen to make them very tidy. Then P goes up and stops, and comes back down to meet Q, which is still on its way up. Writing both positions as "distance below the highest point" makes the collision equation a difference of two squares.

(The printed paper has a misprint in part (i): the "b" in the collision time should be T.)

Solution

We model each particle as a particle on a rough plane, and take positions x measured from A up ℓ, with the positive direction up the slope.



Perpendicular to the plane there is no acceleration, so the normal reaction is

$$R = 4 \cdot 10 \cos 30^\circ = 20\sqrt{3} \text{ N},$$

and since each particle is sliding, friction is limiting:

$$F = \mu R = \frac{1}{5\sqrt{3}} \cdot 20\sqrt{3} = 4 \text{ N}.$$

The weight component down the slope is $40\sin 30^\circ = 20\,\mathrm{N}$.

Moving up: friction acts down the slope, so $4a = -(20+4)$ and the deceleration is $6\,\mathrm{m\,s^{-2}}$.

Moving down: friction acts up the slope, so the acceleration down the slope is $(20-4)/4 = 4\,\mathrm{m\,s^{-2}}$.

Since $20 > 4$, a particle that comes to rest does not stay at rest: the limiting friction cannot hold it, so it slides back down.

Part (i)

Motion of P. Going up, $v = 6-6t$, so P stops at $t=1$, after travelling $\frac{1}{2}\cdot 6\cdot 1 = 3\,\mathrm{m}$. For $0 \le t \le 1$, $x_P = 6t - 3t^2 = 3 - 3(1-t)^2$. Afterwards P slides down from rest at $x=3$ with acceleration 4, so

$$x_P = 3 - 2(t-1)^2 \qquad (t \ge 1),$$

which holds until P returns to A, when $2(t-1)^2 = 3$, that is $t = 1+\sqrt{3/2}$.

Motion of Q. Exactly the same motion shifted by T: with $u = t - T$,

$$x_Q = 6u - 3u^2 = 3 - 3(1-u)^2 \qquad (0 \le u \le 1).$$

Ask yourself Could they collide while both are still going up?

No. While both are going up, Q follows exactly the path P followed T seconds earlier, and x is increasing along that path, so P is strictly ahead. They can only meet once P has turned round, that is for $t \ge 1$. Also Q cannot reach its own highest point ($x=3$, at $u=1$) first, because P has already left $x=3$ by then and is below it. So at the collision P is descending and Q ascending.

The collision. Write $\tau = t - 1 \ge 0$ for the time since P turned. Then $1-u = 1-(t-T) = T-\tau$, and $x_P = x_Q$ gives

$$2\tau^2 = 3(T-\tau)^2.$$

Both τ and $T-\tau = 1-u$ are non-negative (the second because Q is still going up), so we may take positive square roots:

$$\sqrt{2}\,\tau = \sqrt{3}\,(T-\tau)$$

$$\tau = \frac{\sqrt{3}\,T}{\sqrt{3}+\sqrt{2}} = \sqrt{3}(\sqrt{3}-\sqrt{2})\,T = (3-\sqrt{6})\,T,$$

rationalising with $(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2}) = 1$. Hence the collision is at

$$t = (3-\sqrt{6})\,T + 1$$

seconds after P is projected, as required.

It remains to see where the condition on T is used. This time must come before P gets back to A, so we need $\tau < \sqrt{3/2}$:

$$(3-\sqrt{6})\,T < \sqrt{\tfrac{3}{2}} \iff T < \frac{\sqrt{3/2}\,(3+\sqrt{6})}{3} = \sqrt{\tfrac{3}{2}} + \tfrac{\sqrt{9}}{3} = 1 + \sqrt{\tfrac{3}{2}},$$

since $\sqrt{3/2}\cdot\sqrt{6} = 3$. This is exactly the given condition. (The same inequality also guarantees $t \geq T$, so that Q has been launched: $(3-\sqrt{6})T + 1 \geq T \iff T \leq 1/(\sqrt{6}-2) = 1+\sqrt{3/2}$.)

Part (ii)

With $T = 1 + \sqrt{2/3}$,

$$\tau = (3-\sqrt{6})\left(1+\sqrt{\tfrac{2}{3}}\right) = 3 - \sqrt{6} + 3\sqrt{\tfrac{2}{3}} - \sqrt{6}\sqrt{\tfrac{2}{3}} = 3 - \sqrt{6} + \sqrt{6} - 2 = 1,$$

so the collision is at $t = 2$. This value of T is less than $1 + \sqrt{3/2}$, so part (i) applies.

The energy lost to friction is the friction force times the total distance slid, because friction has constant magnitude 4 N and always opposes the motion.

- P slides 3 m up and then $2\tau^2 = 2$ m down: 5 m in all.
- Q has not turned, and is at the same point as P, namely $x = 3 - 2 = 1$: it has slid 1 m.

So the energy lost is

$$4 \times (5+1) = 24 \text{ J}.$$

Checks

An energy balance confirms this. Initially the kinetic energy is $2 \times \frac{1}{2} \cdot 4 \cdot 6^2 = 144$ J. At $t = 2$, P has speed $4 \times 1 = 4$ (KE 32 J); Q has $u = 1 - \sqrt{2/3}$ and speed $6(1-u) = 6\sqrt{2/3}$, so $v^2 = 24$ (KE 48 J). Both are 1 m up the slope, $\frac{1}{2}$ m higher, a gain of $2 \times 4 \cdot 10 \cdot \frac{1}{2} = 40$ J of potential energy. And $144 - 32 - 48 - 40 = 24$ J. A numerical step-by-step simulation of both particles for random $T < 1 + \sqrt{3/2}$ gave collision times matching $(3-\sqrt{6})T + 1$ (for example ≈ 2.000 when $T \approx 1.816$), with 24.000 J lost to friction.

Common slips

- Using the same acceleration for P on the way down as on the way up: friction reverses direction, so it is 4, not 6.
- Forgetting the friction on Q: Q has been sliding too, and its 1 m contributes 4 J.
- Taking the wrong square root in $2\tau^2 = 3(T-\tau)^2$: the sign is fixed by $\tau \geq 0$ and $T - \tau \geq 0$.

Beyond STEP

Friction is the simplest example of a non-conservative force: the work it does depends on the path, not just the endpoints, which is why we needed the total distance slid rather than the displacement. This is why mechanical energy is conserved for some systems and not others, an idea made precise with line integrals and potentials in a first-year mechanics course. Kleppner and Kolenkow's *An Introduction to Mechanics* treats work, conservative forces and friction carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.